Fiscal Year:	FY 2006	**Task Last Updated:**	FY 12/17/2006
PI Name:	Weitz, David Ph.D.		
Project Title:	Fabrication of Photonic Structure with Colloid Engineering		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	COMPLEX FLUIDS/SOFT MATTER--Complex Fluids		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Cambridge	**State:**	MA
Zip Code:	02138	**Congressional District:**	5
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	98-HEDS-03
Start Date:	12/01/2003	**End Date:**	09/30/2006
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA GRC
Contact Monitor:	Griffin, DeVon	**Contact Phone:**	216-433-8109
Contact Email:	DeVon.Griffin@grc.nasa.gov , DeVon.W.Griffin@nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	0		
COI Name (Institution):			
Grant/Contract No.:	NNC04GA55G		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The goal of this work is to investigate the growth and properties of advanced photonic materials using a microgravity environment. The work focuses on photonic devices created by colloidal engineering using a combination of new colloidal particles and highly organized and mixed structures. The goal of the experiments is to create novel photonic devices using colloidal particles as precursors. To achieve this goal, we will work simultaneously on several different components of the research. We will investigate new methods for synthesizing the colloidal particles themselves, which serve as the precursors for all the structures we will create. Our goal is to improve the liquid crystal emulsion droplets to achieve better switching properties and to incorporate new materials that will improve the optical behavior of the final devices.		

Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	This project was ended early and the final installment of the support was reduced. Thus, the work on the project was not finished by the end of the grant period. We were working on a new type of photonic device, fabricated using microfluidic techniques. We successfully fabricated shells of liquid crystals, with water on the inside and water on the outside, separated by a shell of liquid crystal. These structures have fascinating defect behavior, and we have investigated this behavior as a function of the thickness of the shell. The results are in good accord with theory. To demonstrate the utility and photonic activity of these particles, we used laser tweezers with rotationally polarized light. This both traps the particles and causes them to rotate, thus demonstrating the optical anisotropy and activity. A complete report has been published in Advanced Materials.
Bibliography Type:	Description: (Last Updated: 06/13/2025)
Articles in Peer-reviewed Journals	Fernández-Nieves A, Cristobal G, Garcés-Chávez V, Spalding GC, Dholakia K, Weitz DA. "Optically anisotropic colloids of controllable shape." Advanced Materials 2005 Mar;17(6):680-4. http://dx.doi.org/10.1002/adma.200401462 , Mar-2005
Articles in Peer-reviewed Journals	Fernández-Nieves A, Vitelli V, Utada AS, Link DR, Márquez M, Nelson DR, Weitz DA. "Novel defect structures in nematic liquid crystal shells." Phys Rev Lett. 2007 Oct 12;99(15):157801. PMID: 17995213 , Oct-2007